

The Dimension of Nonterminating Resampling Computations

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Abstract

A randomized algorithm can terminate almost surely yet run forever on exceptional random tapes. This paper studies the tail probability of continued execution, the algorithmic randomness of one such tape, and the Hausdorff dimension of the whole exceptional set. Reversing any finite variable-model resampling run yields an exact likelihood ratio. Its information surplus Δ_T satisfies $\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}$; for computable instances, the same identity bounds the Kolmogorov complexity of every surviving tape prefix. Thus running time and pointwise complexity are controlled by one comparison.

Exact answers are obtained in two local-repair regimes. For extremal partial rejection sampling, a source-power extension of Cartier–Foata trace enumeration places the nontermination dimension at the Shearer boundary. For nonextremal disagreement repair, Hamming weight determines the complete survival law, the Hausdorff dimension, and a dimension-attaining change of measure, independently of the connected graph’s topology and the deterministic nonanticipating repair rule. Under computability it also gives the maximal effective strong dimension. The examples also show that dependency graphs and event marginals need not determine exact survival or dimension, and that a state sufficient for these quantities may be exponentially smaller than one sufficient to continue the computation. This paper does not find such a small exact state for arbitrary overlapping repairs.

1 Introduction

Randomized algorithms are usually studied by averaging over their random tapes. Nontermination can instead be studied one tape at a time. Once the tape is fixed, the computation is deterministic: it either stops or it does not. Let $\mathbb{P}$ denote the source law of the tape and put

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where τ is the number of rejection or repair steps before stopping. The tail $\mathbb{P}\{\tau \geq T\}$ measures how often the computation survives T steps. When the following extended limit exists, the *survival exponent* (or base-two escape rate) is

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This terminology is standard for exponential escape from a surviving region (Keller and Liverani, 2009). This distributional quantity does not describe the exceptional tapes themselves. Even when $\mathbb{P}(\mathcal{N}) = 0$, the set $\mathcal{N}$ may be nonempty and may contain incompressible sequences. Effective dimension measures the algorithmic randomness of one fixed tape; source-relative Hausdorff dimension measures the size of the whole nonterminating set.

The Lovász local lemma has a different scope. It gives conditions under which all bad events can be avoided, and its algorithmic forms give termination or tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris, 2017). The present paper fixes the resampling computation and asks for its exact survival decay and for the complexity and geometry of its exceptional tapes. These are questions about a specified process, not a stronger universal local-lemma criterion.

1.1 Motivation from pointwise learning and localization

In statistical learning, $\hat{h}(S)$ depends on the training sample S , so a concentration inequality proved for one fixed hypothesis cannot simply be evaluated at $h = \hat{h}(S)$. Uniform convergence handles this data-dependent choice by controlling a class simultaneously; localized empirical-process methods sharpen the control by restricting attention to a region relevant to the realized estimator (Bartlett et al., 2005; Koltchinskii, 2006).

A resampling computation has an analogous dependence between a random tape and the repair history that it generates. The event $\{\tau \geq T\}$ restricts tape space to the cylinders generated by the minimal live prefixes $\mathcal{L}_T$. These events are nested, with $\{\tau \geq T + 1\} \subseteq \{\tau \geq T\}$ and $\mathcal{N} = \bigcap_T \{\tau \geq T\}$. The backward likelihood ratio below studies one realized member of this family, whereas the source-power sum studies the family as a whole.

Recent work studies complexity localized at an individual trained network or at one index of a Gaussian field (Li and Xu, 2026; Xu, 2026b). Here the individual object is a random tape. The later search for a smaller state that preserves specified quantities also parallels decision-sufficient state compression, termed Bellman sufficiency in Xu (2026a).

	Statistical learning	Resampling computation
Random input	training sample S	random tape ω
Selected object	learned hypothesis $\hat{h}(S)$	repair history and live prefix selected by ω
Localized set	hypotheses in a relevant data-dependent region	prefixes $\mathcal{L}_T$ that produce at least T repairs
Set-level control	localized empirical-process or Rademacher complexity	weighted mass of the tape cylinders that survive T repairs
Question	generalization of the selected hypothesis	survival and the dimensions of nontermination

Table 1: A structural analogy between localization in learning and the tape-level analysis of resampling. It is a motivation, not a reduction.

The analogy concerns data-dependent localization, not an identification of complexity notions. Local Rademacher complexity, prefixwise Kolmogorov complexity, and weighted cylinder sums are different mathematical objects, and the results below do not imply a learning-theoretic generalization bound. The common viewpoint is to measure the region retained by an adaptive interaction rather than return immediately to a worst-case bound over the ambient space.

1.2 Results and scope

Two objects organize the paper. First, reversing a finite variable-model resampling run gives an exact likelihood ratio. The sum of the repaired events’ marginal surprisals, minus the information

needed to describe their order, has an exponential upper tail. Under computability, a code for the same backward law shows that a positive surplus shortens the description of the tape prefix by that amount, up to an additive constant. This identity holds without a local-lemma hypothesis.

Second, for the set $\mathcal{L}_T$ of prefixes that remain live for T repairs, define

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s.$$

This is the usual Rényi partition sum of the live cylinders; its logarithmic exponential growth is pressure (Rényi, 1961; Bowen, 1975). The term *source-power sum* is used only as shorthand for this standard object. At $s = 0$ this counts feasible prefixes, and at $s = 1$ it is the survival probability. Intermediate powers balance the growth of possible runs against the source information needed to realize them. In the exact regimes below, the growth of $Z_T(1)$ gives the survival exponent, the critical power gives the Hausdorff dimension, and a dimension-attaining change of measure produces tapes attaining the pointwise effective-dimension bound under computability.

The principal results are as follows.

1. Theorem 3.1 is a finite-run identity for every variable-model resampling computation. It yields a universal tail bound, an individual-tape coding bound, and an exact relative-entropy expression for the slack.
2. Theorem 5.2 solves every finite extremal partial-rejection instance. Powering each consumed source value lifts the classical Cartier–Foata trace series to the complete source-power family; the nontermination dimension is the value at which the source-power event masses reach the Shearer boundary.
3. Theorem 5.5 solves a genuinely nonextremal family. For disagreement repair on any connected graph, Hamming weight preserves every source-power sum, independently of the deterministic nonanticipating repair rule. On a path under the leftmost rule, the resulting pressure calculation uses linearly many states, whereas every self-contained exact continuation state has exponentially many values.

Several ingredients are classical: effective and source-relative dimension (Billingsley, 1965; Athreya et al., 2007; Lutz, 2011), finite-state pressure (Parry, 1964; Mauldin and Williams, 1988; Bowen, 1975; Simpson, 2015), reversible entropy-compression records (Moser, 2009; Moser and Tardos, 2010), and the Cartier–Foata representation of extremal partial rejection (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). The new step is the exact backward likelihood formulation and its joint tail and coding consequences, the source-power extension of the trace formula for finite extremal instances, and the all-power Hamming-weight reduction for nonextremal disagreement repair.

The limits are equally important. Whole-block systems with the same dependency graph and event marginals can have different survival exponents and dimensions because the graph forgets event intersections. Conversely, when the selector is determined by the current assignment, that assignment gives an exact finite-state operator; any finite controller memory can be adjoined to it. This exact state space is usually exponential. For arbitrary nonextremal overlapping repairs, this paper does not find a comparably small exact state description. Section 2 begins with rejection sampling; Sections 3–5 develop the likelihood identity, the graph separation, and the two exact local-repair regimes. The main text ends with the conclusions, and all proofs are collected in the appendices.

2 Rejection sampling: an exact warm-up

A familiar example is the simplest form of rejection sampling. Draw $X_1, X_2, \dots$ independently from a full-support probability law π on a finite set Ω with $|\Omega| \geq 2$, reject a draw in $U \subseteq \Omega$, and return the first draw in $G = \Omega \setminus U$. This is the special case of the classical accept–reject method in which each proposed value is accepted or rejected deterministically (von Neumann, 1951; Devroye, 1986).

Write $p = \pi(U)$. If $0 < p < 1$, then

$$\mathbb{P}\{\tau \geq n\} = p^n, \quad \mathbb{P}\{\tau = t\} = p^t(1 - p), \quad \mathbb{E}\tau = \frac{p}{1 - p} \quad (n, t \in \mathbb{Z}_{\geq 0}). \quad (1)$$

Here τ is the number of rejected draws; the returned value has law $\pi(\cdot \mid G)$. If $p = 0$, the first draw is accepted; if $p = 1$, the algorithm never stops. When $0 < p < 1$, the algorithm stops almost surely, although the set $U^{\mathbb{N}}$ of tapes on which it never stops is nonempty. Probability assigns this set mass zero. Effective dimension asks how much algorithmic randomness can nevertheless remain in one of its individual elements. The geometric survival law in (1) is standard; its role here is to provide the smallest case in which survival, individual-tape complexity, and set dimension can all be computed exactly.

First suppose that the source is uniform and $U \neq \emptyset$. Write $q = |\Omega| \geq 2$ and $M = |U|$, so $1 \leq M \leq q$. The probability that the first n blocks are rejected is

$$\left(\frac{M}{q}\right)^n,$$

and there are exactly M^n possible surviving prefixes. Each can be specified using $n \log_2 M + O(\log n)$ bits, whereas an unrestricted source prefix contains $n \log_2 q$ bits. The ratio is

$$\frac{\log_2 M}{\log_2 q}.$$

In the two-bit example

$$\Omega = \{00, 01, 10, 11\}, \quad U = \{00, 01\}.$$

Every surviving block has its first bit fixed and its second bit free. At most half of the source information can remain, and suitable tapes attain this value.

For a computable full-support source $P = \pi^{\mathbb{N}}$, let $\omega_{1:n}$ denote the first n source symbols (or blocks). For a finite word w , let $[w]$ be the cylinder of tapes beginning with w , and write $P[w] = P([w])$. Define their self-information by

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let $K(w)$ be the length of the shortest program that prints w on a fixed universal prefix-free, or self-delimiting, machine (Li and Vitányi, 2019). More generally, $K(w \mid z)$ is the corresponding program length when z is supplied as auxiliary input. The effective strong P -dimension of a tape ω is

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \quad (2)$$

Replacing the limsup by a liminf defines the effective P -dimension $\dim_{\text{eff}, P}(\omega)$. For a fair binary source, write

$$\text{Dim}_{\text{eff}}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{n}.$$

This is the standard constructive strong dimension relative to a source measure (Athreya et al., 2007; Lutz, 2011); the definition itself is not new. It is used here to measure the algorithmic randomness of one realized tape, in the pointwise sense discussed in Section 1.1. A value near one means that the tape is nearly as incompressible as an ordinary draw from P , whereas a smaller value means that continued execution has imposed additional structure on it.

For a set E of tapes, $\dim_H^P E$ denotes its Billingsley, or source-relative Hausdorff, dimension: in the Hausdorff covering construction, the cylinder $[w]$ is assigned diameter $P[w]$ (Billingsley, 1965; Lutz, 2011). Under a uniform q -ary source, this is ordinary Hausdorff dimension in the symbolic metric $d(\omega, \eta) = q^{-k}$, where k is the length of the longest common prefix of ω and η .

The three quantities answer different questions:

- (i) $\mathbb{P}\{\tau \geq n\}$ is the probability of performing at least n rejection or repair steps;
- (ii) $\text{Dim}_{\text{eff}, P}(\omega)$ measures the randomness of one tape; and
- (iii) $\dim_H^P \mathcal{N}$ measures the size of the set of all nonterminating tapes.

None of these notions is introduced here. The question is whether one fixed resampling computation links them and when all three can be computed exactly. The first is distributional, the second is pointwise, and the third is geometric. In general, effective strong dimension, which uses a limsup, is associated with packing dimension rather than Hausdorff dimension (Athreya et al., 2007). The equality between maximal effective strong dimension and Hausdorff dimension proved below is a feature of the regular product and finite-state sets considered here, not a general identity. For rejection sampling, the tail in (i) is $\pi(U)^n$. When U is nonempty, the two dimension questions are governed by the number δ that solves

$$\sum_{x \in U} \pi(x)^\delta = 1. \quad (3)$$

Every tape rejected forever then satisfies

$$\text{Dim}_{\text{eff}, P}(\omega) \leq \delta \quad (\omega \in U^\mathbb{N}). \quad (4)$$

To see that the bound is attained, write $\mathbf{1}\{\cdot\}$ for the indicator of the stated condition and set

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}.$$

For a computable law Q , a tape is Martin–Löf random if it avoids every effective Q -null test; equivalently, the Levin–Schnorr theorem gives $K(\omega_{1:n}) \geq -\log_2 Q[\omega_{1:n}] - O(1)$ for every n (Li and Vitányi, 2019). The displayed Q_δ is a probability law. A tape that is Martin–Löf random under $Q_\delta^\mathbb{N}$ attains equality. The corresponding Hausdorff calculation gives $\dim_H^P(U^\mathbb{N}) = \delta$. Under the uniform law, $\pi(U) = q^{\delta-1}$, so the tail exponent and the dimension determine one another. For a nonuniform law, they need not.

The following bounds show what remains true without uniformity. Let $p = \pi(U)$, let $\nu = \pi(\cdot \mid U)$ be the law of a rejected symbol, and write $\alpha = \log_2(1/p)$ for the survival exponent.

Proposition 2.1 (Entropy bounds for the rejection dimension). *Assume $0 < p < 1$. If $|U| = 1$, then $\delta = 0$. If $|U| \geq 2$, define the Rényi entropy of order $s \in (0, 1)$ by*

$$H_s(\nu) = \frac{1}{1-s} \log_2 \sum_{x \in U} \nu(x)^s,$$

and let $H(\nu)$ be the Shannon entropy (Rényi, 1961; Shannon, 1948). Then

$$\delta = \frac{H_\delta(\nu)}{\alpha + H_\delta(\nu)} \quad (5)$$

and hence

$$\frac{H(\nu)}{\alpha + H(\nu)} \leq \delta \leq \frac{\log_2 |U|}{\alpha + \log_2 |U|}. \quad (6)$$

The two bounds coincide when ν is uniform on U .

The proof is given in Appendix B.

Thus α is the information cost of surviving one more draw, while $H_\delta(\nu)$ measures the choice that remains among rejected symbols. Their balance determines the dimension. The law Q_δ produces tapes attaining the pointwise bound, and regularity of the product set makes the Hausdorff dimension equal to the same value.

3 A backward likelihood identity

Let $X_1, \dots, X_n$ be independent finite-valued variables with product law $\pi = \bigotimes_i \pi_i$. For $J \subseteq \{1, \dots, n\}$, write $\pi_J = \bigotimes_{i \in J} \pi_i$ for the marginal product law on the coordinates in J . Each nonempty bad event A is determined by the coordinates $v(A)$ and has probability $p_A > 0$. Its *marginal surprisal* is $\log_2(1/p_A)$. Fix a deterministic rule for selecting a currently true bad event.

List the random choices in the order in which the algorithm uses them. First draw $S_0 \sim \pi$. Whenever the current assignment contains a bad event, select A_t , draw $Z_t \sim \pi_{v(A_t)}$ independently of the past, and write Z_t on $v(A_t)$. Let S_t denote the resulting assignment after the t th repair. Let E_T be the event that at least T resamplings occur, and put

$$\Xi_T = (S_0, Z_1, \dots, Z_T).$$

Let P_{tape} denote the law generated by the initial draw and these successive independent resampling draws. Then $\mathbb{P}(E) = P_{\text{tape}}(E)$ for every tape event E ; for a finite used-tape prefix w (including Ξ_T), $P_{\text{tape}}(w)$ denotes the probability of its cylinder. Thus $E_T = \{\tau \geq T\}$. For each fixed T , the minimal tape prefixes that realize T resamplings form a prefix-free family: no member is a proper prefix of another. Their cylinders are therefore disjoint, and their probabilities sum to $\mathbb{P}(E_T)$.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the configuration overwritten at time t , and let $L_T = (A_1, \dots, A_T)$. Write $\pi_{v(A)}(\cdot \mid A)$ for the marginal law $\pi_{v(A)}$ conditioned on the local configurations satisfying A . Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t \mid A_t). \quad (7)$$

Let Q_T be a subprobability law on repair histories $\ell = (A_1, \dots, A_T)$, positive on every feasible history of length T . Thus $-\log_2 Q_T(\ell)$ is its ideal code length. Define the backward subprobability law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t \mid A_t). \quad (8)$$

The map

$$\Xi_T \mapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from T to 1: the current values on $v(A_t)$ recover Z_t , and restoring Y_t recovers the preceding assignment.

Theorem 3.1 (Backward likelihood identity). *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (9)$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (10)$$

If the instance and the family $(Q_T)_{T \geq 1}$ are uniformly computable, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T \mid T, \text{instance}) \geq \Delta_T - O(1). \quad (11)$$

Here instance denotes a fixed effective description of the source laws, bad events, and deterministic selection rule. Uniform computability means that one algorithm can approximate $Q_T(\ell)$ to any requested precision from T and ℓ , with the instance data encoded effectively in the same sense.

For use in the exact slack formula below, let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes producing at least T repairs and write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)), \quad \mu_T(w) = G_T(R_T(w)).$$

Backward reconstruction makes R_T injective. Since G_T has total mass at most one, μ_T is a subprobability mass on $\mathcal{W}_T$.

The proof is given in Appendix C.

The likelihood identity also identifies the exact slack after conditioning on a long run. Let $p_T = P_{\text{tape}}(E_T) > 0$, let $m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w) > 0$, and define probability laws on $\mathcal{W}_T$ by

$$P_T(w) = \frac{P_{\text{tape}}(w)}{p_T}, \quad \nu_T(w) = \frac{\mu_T(w)}{m_T}.$$

Write D_2 for relative entropy in bits and H_2 for Shannon entropy in bits.

Corollary 3.2 (Exact slack and the optimal history law). *For every fixed comparison law Q_T ,*

$$-\log_2 p_T = \mathbb{E}_{P_T} \Delta_T + D_2(P_T \parallel \nu_T) - \log_2 m_T. \quad (12)$$

Moreover, if $S_T = \sum_{t=1}^T \log_2(1/p_{A_t})$, then

$$\sup_{Q_T} \mathbb{E}[\Delta_T \mid E_T] = \mathbb{E}[S_T \mid E_T] - H_2(L_T \mid E_T), \quad (13)$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by $Q_T(\ell) = \mathbb{P}\{L_T = \ell \mid E_T\}$ on the conditional support.

The proof is given in Appendix C.

Thus the difference between the escape information $-\log_2 p_T$ and the mean surplus consists exactly of a mismatch term and the unused backward mass. The theorem also gives the normalized bound

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} \leq 2^{-aT} \quad (T \geq 1, a \geq 0). \quad (14)$$

Corollary 3.3 (Sharpness of the surplus tail). *Neither the constant nor the exponent in (14) can be improved universally. For any $a > 0$, a one-event rejection sampler with event probability $p = 2^{-a}$ and its unique repair history assigned probability one satisfies, for every T ,*

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} = \mathbb{P}(E_T) = p^T = 2^{-aT}. \quad (15)$$

Large-deviation interpretation. No large-deviation principle is assumed in the paper. If Δ_T/T is known separately to satisfy one with a base-two rate function J , and the principle identifies the exact upper-tail rate at a , then

$$\inf_{x \geq a} J(x) \geq a. \quad (16)$$

The one-event example shows that equality is possible. For ordinary rejection sampling with rejection probability p , survival means that T independent Bernoulli variables all equal one, so its rate is the endpoint relative entropy

$$D_2(1||p) = \log_2 \frac{1}{p}.$$

For the finite-state model in Appendix A, the corresponding escape rate is $-\log_2 \rho(B_1)$ (Dembo and Zeitouni, 2010; Keller and Liverani, 2009).

The two terms in (9) have different origins. The first sums the marginal surprisals of the events that were repaired; the second is the ideal code length of their ordered history. The first sum may count statistically dependent information more than once. Neither term alone gives a bound. Their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with q colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability $1/q$, so each repair contributes $\log_2 q$ bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

Corollary 3.4 (A tail bound and a pointwise dimension bound). *Suppose every T -repair execution satisfies*

$$\sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \quad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where $\beta > \gamma$. Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta - \gamma)T}. \quad (17)$$

For the pointwise conclusion, assume in addition that

- (i) the instance and the family $(Q_T)_{T \geq 1}$ are uniformly computable;
- (ii) the computation reads a fixed fair binary tape sequentially, the initial draw consumes $c_0 = O(1)$ bits, and every repair consumes exactly c bits, so that Ξ_T is uniformly computably interconvertible with $\omega_{1:c_0+cT}$.

Then every nonterminating tape satisfies

$$\text{Dim}_{\text{eff}}(\omega) \leq \max \left\{ 0, 1 - \frac{\beta - \gamma}{c} \right\}. \quad (18)$$

The proof is given in Appendix C.

Remark 3.5 (What the record must contain). For a recursive Moser-style implementation, L_T must be enlarged to include the recursion tree or return structure unless those data are recoverable by a separate prefix-free decoding rule. In Theorem 3.1, everything needed for reversal contributes to the probability $Q_T(L_T)$.

4 What graph summaries omit

To use Theorem 3.1, one must describe the possible lists of repairs. The Lovász local lemma summarizes event interactions by a dependency graph, and many algorithmic proofs use that graph to organize repair histories. This section asks what those graph data determine about the exact tail and the nonterminating tapes of a fixed computation.

4.1 Why dependency graphs enter

Let $\mathcal{A}$ be a finite family of bad events. The Lovász local lemma asks when there is a positive probability of avoiding all of them:

$$\mathbb{P}\left(\bigcap_{A \in \mathcal{A}} A^c\right) > 0. \quad (19)$$

In the variable model, where independent variables generate the events, the standard graph D joins A and B whenever their variable sets overlap. For example, in random vertex coloring a bad event says that one edge is monochromatic, and two such events are adjacent in D when their edges share a vertex. More generally, write $\Gamma(A)$ for the neighbors of A . A loopless graph on $\mathcal{A}$ is a dependency graph if every event A is independent of the sigma-field generated by its nonneighbors; equivalently, A is independent of every Boolean combination of events indexed by $\mathcal{A} \setminus (\{A\} \cup \Gamma(A))$. The vertices of D are events; this is different from the bipartite event–variable incidence graph used in Section 5. If $\mathbb{P}(A) \leq p$ for every A and D has maximum degree d , the symmetric local lemma gives the familiar sufficient condition

$$ep(d+1) \leq 1. \quad (20)$$

Here e is Euler’s number. For a fixed graph D , Shearer’s region is the set of probability vectors $(p_A)_{A \in \mathcal{A}}$ for which every event family having dependency graph D and satisfying $\mathbb{P}(A) \leq p_A$ has positive probability of avoiding all events. It is the sharp worst-case criterion based only on D and the marginal event probabilities (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005).

Moser and Tardos turn this existence argument into a random process. Draw all variables independently; while some bad event is true, choose one and redraw the variables on which it depends. If there are numbers $x_A \in (0, 1)$ such that

$$\mathbb{P}(A) \leq x_A \prod_{B \in \Gamma(A)} (1 - x_B) \quad (A \in \mathcal{A}), \quad (21)$$

then the expected number of times A is redrawn is at most $x_A/(1 - x_A)$ (Moser and Tardos, 2010). Consequently,

$$\mathbb{E}\tau \leq \sum_{A \in \mathcal{A}} \frac{x_A}{1 - x_A} < \infty. \quad (22)$$

The algorithm therefore stops almost surely and returns an assignment in which no bad event occurs. Bounds of this kind are proved by encoding repair histories as trees, directed acyclic graphs (DAGs), or entropy-compression records whose weights are determined by event probabilities and adjacency in D (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2022).

The graph summary omits intersection data. The graph records which repairs may interact, but it does not determine $\mathbb{P}(A \cap B)$, set inclusion, or $\mathbb{P}(\bigcup_A A)$. Two event systems can therefore have the same graph and the same marginal probabilities but different survival exponents and different nonterminating sets. The remainder of the section compares the exact state-space description, the graph bound, and whole-block examples in which the loss can be computed explicitly.

4.2 The full state space

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Set $P = P_{\text{tape}}$, the sequential source law of the initial assignment and subsequent resampling draws; finite initial factors do not affect the dimensions below. Delete zero-probability outcomes and assume that every surviving source outcome has probability strictly less than one. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments and for $\theta \geq 0$, define

$$(B_\theta)_{xy} = \sum_{z: F(x,z)=y} \pi_{v(A(x))}(z)^\theta, \quad (23)$$

where $A(x)$ is the selected event and $F(x, z)$ is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components C containing a directed cycle, where a strongly connected component is a maximal set of states that are mutually reachable by directed paths. Let $B_{\theta,C}$ be the restriction of B_θ to C . If no such component exists, then $\mathcal{N} = \emptyset$ and its dimension is zero. Otherwise, Proposition A.1 gives

$$\dim_H^P \mathcal{N} = \max_C \delta_C, \quad \rho(B_{\delta_C, C}) = 1. \quad (24)$$

Here $\delta_C \in [0, 1]$ is the unique solution of the displayed spectral-radius equation for component C . This formula is exact, but its matrix usually has exponentially many rows and columns.

4.3 The graph bound

Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities p_A to bound the total weight of possible repair lists (Cartier and Foata, 1969; Scott and Sokal, 2005). Only the complete-graph case is needed here. If $D = K_m$ is complete on $m = |\mathcal{A}|$ event vertices, the graph-only enumeration allows every word $(A_1, \dots, A_t) \in \mathcal{A}^t$ and assigns it weight $\prod_{i=1}^t p_{A_i}$. Its total weight is

$$\left(\sum_{A \in \mathcal{A}} p_A \right)^t.$$

Put $S = \sum_A p_A$ and choose the history law

$$Q_T(A_1, \dots, A_T) = \frac{\prod_{t=1}^T p_{A_t}}{S^T}$$

on all event words. Theorem 3.1 then gives $\Delta_T = -T \log_2 S$. If $S < 1$, it follows that

$$\mathbb{P}(E_T) \leq S^T. \quad (25)$$

For a computable uniform q -ary block instance of size b , let P be the product source on blocks. Summing the compatible overwritten blocks over all event words gives at most

$$\sum_{(A_1, \dots, A_T)} \prod_{t=1}^T |A_t| = (q^b S)^T$$

length- T overwritten-block strings. The final block contributes only the fixed factor q^b , which disappears after normalization. Coding the resulting prefixes gives

$$d_{\text{graph}} = \left\lceil 1 + \frac{\log_2 S}{b \log_2 q} \right\rceil_{[0,1]}, \quad \text{Dim}_{\text{eff},P}(\omega) \leq d_{\text{graph}} \quad (\omega \in \mathcal{N}), \quad \dim_H^P \mathcal{N} \leq d_{\text{graph}}, \quad (26)$$

where $[\cdot]_{[0,1]}$ denotes truncation to $[0, 1]$. These are graph-based upper bounds; they need not equal the survival probability or dimension for a fixed selection rule. A short graph description also does not make exact computation easy: evaluating the relevant independence polynomial and testing membership in the general Shearer region are $\#P$ -hard in general (Harvey et al., 2018).

4.4 Whole-block resampling

Let Ω be a finite block space with source law π satisfying $0 < \pi(x) < 1$ for every x , and let $\mathcal{A}$ be any finite family of bad subsets. Every repair redraws the whole block from π . Fix a deterministic selector ϕ that, for each $x \in U = \bigcup_{A \in \mathcal{A}} A$, chooses one event $\phi(x)$ containing x . The selector partitions the bad union into the disjoint regions

$$C_A = \{x \in U : \phi(x) = A\}. \quad (27)$$

Let $\mathcal{I} = \{A \in \mathcal{A} : C_A \neq \emptyset\}$ index the nonempty regions. For a laminar family with the inclusion-minimal selector, C_A is A minus its maximal proper subevents. Define

$$w_A(\theta) = \sum_{x \in C_A} \pi(x)^\theta, \quad W(\theta) = \sum_{A \in \mathcal{I}} w_A(\theta) = \sum_{x \in U} \pi(x)^\theta. \quad (28)$$

Write $w(\theta) = (w_A(\theta))_{A \in \mathcal{I}}$ for the corresponding column vector.

Theorem 4.1 (Exact whole-block calculation). *Assume $U \neq \emptyset$, and let $P = \pi^{\mathbb{N}}$. There is a unique $\delta \in [0, 1]$ such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^\delta = 1. \quad (29)$$

(a) *If the alphabet, π , U , and the law Q_δ below are computable, every nonterminating tape satisfies*

$$\text{Dim}_{\text{eff},P}(\omega) \leq \delta \quad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \quad (30)$$

Every tape that is Martin–Löf random for $Q_\delta^{\mathbb{N}}$ satisfies $\dim_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$. Moreover,

$$\dim_H^P \mathcal{N} = \delta, \quad \mathbb{P}\{\tau \geq t\} = \pi(U)^t. \quad (31)$$

Under the uniform law on $\Omega = [q]^b$, writing $M = |U|$,

$$\dim_H \mathcal{N} = \delta = \frac{\log_2 M}{b \log_2 q}, \quad \mathbb{P}\{\tau \geq t\} = \left(\frac{M}{q^b}\right)^t. \quad (32)$$

In particular, under a uniform source, a tape of effective strong dimension greater than δ must make the algorithm terminate.

(b) *Group rejected blocks according to the region C_A that contains them. For every $\theta \geq 0$, the resulting weighted transition matrix is*

$$\widehat{B}_\theta(A, B) = w_B(\theta) \quad (A, B \in \mathcal{I}). \quad (33)$$

Its spectral radius is $W(\theta)$, the same as that of the full transition matrix, and the all-ones vector $\mathbf{1} \in \mathbb{R}^{|\mathcal{I}|}$ is a right eigenvector.

(c) The dimension and survival bounds are attained by the following reversible record laws. The dimension-attaining law is

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}. \quad (34)$$

For every T -repair execution, there is a probability record law $G_{\delta,T}$ satisfying

$$\log_2 \frac{G_{\delta,T}(S_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = (1 - \delta) \sum_{t=1}^T \log_2 \frac{1}{\pi(Y_t)}. \quad (35)$$

The second is the source conditioned on another rejection, $Q_{\text{surv}}(x) = \pi(x) \mathbf{1}\{x \in U\} / \pi(U)$. The corresponding record law $G_{\text{surv},T}$ satisfies

$$\log_2 \frac{G_{\text{surv},T}(S_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = T \log_2 \frac{1}{\pi(U)}. \quad (36)$$

The proof is given in Appendix D.

Remark 4.2 (The one-state reduction). Every repair redraws the whole block. Consequently, the next step is independent of the current rejected block, and the one-state matrix $[W(\theta)]$ already gives both $W(1) = \pi(U)$ and the dimension root. The disjoint regions retain the recorded reason for rejection, but do not change the stopping dynamics. Partially overlapping local repairs, which leave relevant state unchanged, are the harder case.

The survival law is also a direct application of Theorem 3.1. Relabel each repair by the disjoint region C_A containing the overwritten block; this does not change the run or its resampling rule. Put $r_A = \pi(C_A)$ and choose

$$Q_T(A_1, \dots, A_T) = \prod_{t=1}^T \frac{r_{A_t}}{\pi(U)}.$$

For the relabeled event C_{A_t} , the first term in (9) is $\log_2(1/r_{A_t})$, whereas encoding the region label under Q_T costs $\log_2(\pi(U)/r_{A_t})$ bits. Their difference is $\log_2(1/\pi(U))$ at every repair, giving (36). Conditioning only on the original event A_t would count overlapping mass more than once.

Comparison with the dependency graph. Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight $\sum_{A \in \mathcal{A}} \pi(A)$, while the true rejection probability is $\pi(U)$. Under the uniform source on $[q]^b$, the resulting upper bound and the true dimension are

$$1 + \frac{\log_2 \sum_{A \in \mathcal{A}} \pi(A)}{b \log_2 q} \quad \text{and} \quad 1 + \frac{\log_2 \pi(U)}{b \log_2 q}, \quad (37)$$

again truncated to $[0, 1]$. Before truncation, the two expressions agree when the events are disjoint, whereas overlap produces overcounting. Truncation can conceal this difference when both values are at least one.

In the variable model, a nonedge certifies disjoint scopes and hence commuting repairs; an edge marks only possible interaction. A complete graph therefore contains no information about how the corresponding bad events intersect. The following standard family makes the loss explicit.

Example 4.3 (Nested Hamming balls). Let $b \geq 1$ and $0 \leq r \leq b$ be integers, and let $\Omega = \{0, 1\}^b$ carry the uniform law. Reject a block when its Hamming weight is at most r . Writing $|x|$ for the Hamming weight of x , let

$$U_r = \{x : |x| \leq r\}, \quad M_r = |U_r| = \sum_{j=0}^r \binom{b}{j}.$$

Theorem 4.1 gives

$$\text{Dim}_{\text{eff}}(\omega) \leq \frac{1}{b} \log_2 M_r \quad \text{for every } \omega \in U_r^{\mathbb{N}}, \quad \dim_H(U_r^{\mathbb{N}}) = \frac{1}{b} \log_2 M_r. \quad (38)$$

Now add the stricter rule $U_s \subset U_r$, where $0 \leq s < r$, and record U_s whenever it applies. The standard variable-overlap dependency graph on the two original rules is K_2 and the graph-only sum is $(M_s + M_r)/2^b$, although the computation rejects on a set of mass only $M_r/2^b$.

For example, take $b \geq 3$, $r = b - 1$, and $s = b - 2$. The graph sum is $2 - (b + 2)/2^b > 1$, so the graph bound gives only the trivial upper bound 1, whereas the true dimension is $b^{-1} \log_2(2^b - 1) < 1$. Splitting U_r into U_s and $U_r \setminus U_s$ removes this double counting.

Example 4.4 (Same local-lemma data, different survival exponents). The graph and bad-event probabilities also fail to determine the exact survival exponent and dimensions. Let π be uniform on $\Omega = \{0, 1\}^3$, and define

$$\begin{aligned} B &= \{000, 100, 010, 001\}, \\ A &= \{000, 111, 110\}, \\ A' &= \{000, 100, 111\}. \end{aligned}$$

Membership in each event depends essentially on all three coordinates. Consequently, the systems $\{A, B\}$ and $\{A', B\}$ have the same event-variable incidence graph $K_{2,3}$ and the same dependency graph K_2 . Their ordered vectors of event probabilities are also identical:

$$(\pi(A), \pi(B)) = (\pi(A'), \pi(B)) = \left(\frac{3}{8}, \frac{1}{2}\right).$$

What differs is the intersection: $|A \cap B| = 1$ but $|A' \cap B| = 2$. Neither event contains the other. Hence

$$|A \cup B| = 6, \quad |A' \cup B| = 5.$$

If every repair redraws the whole block, the two survival probabilities are

$$\left(\frac{3}{4}\right)^t \quad \text{and} \quad \left(\frac{5}{8}\right)^t,$$

with survival exponents

$$\alpha_1 = \log_2 \frac{4}{3}, \quad \alpha_2 = \log_2 \frac{8}{5}.$$

Let $\mathcal{N}_1 = (A \cup B)^{\mathbb{N}}$ and $\mathcal{N}_2 = (A' \cup B)^{\mathbb{N}}$. Theorem 4.1 gives

$$\max_{\omega \in \mathcal{N}_1} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_1 = \frac{\log_2 6}{3}, \quad \max_{\omega \in \mathcal{N}_2} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_2 = \frac{\log_2 5}{3}.$$

Thus the common graph and bad-event probabilities do not determine the exact survival exponent and, consequently in this uniform whole-block setting, do not determine the maximal effective strong dimension or the Hausdorff dimension.

Relation to the variable local lemma. The separation does not contradict the tightness of the Shearer region, which is a worst-case statement over event systems with fixed graph and event probabilities. In fact, the common marginal vector in Example 4.4 satisfies

$$\pi(A) + \pi(B) = \frac{7}{8} < 1,$$

and therefore lies strictly inside the Shearer region for K_2 . The variable local lemma gives avoidability guarantees from event probabilities and event-variable structure (He et al., 2017); algorithmic refinements give convergence bounds, including criteria that use intersection information (Moser and Tardos, 2010; He et al., 2023). The present questions have a different output. The backward identity supplies tail and individual-tape complexity bounds for every finite run in the model, whether or not a local-lemma criterion is satisfied. For the whole-block systems solved here, combining this identity with the product-set calculation determines the exact survival exponent and both dimensions. What differs in Example 4.4 is not local-lemma feasibility but the exact survival law and exceptional set of the specified whole-block process. This is a tape-level analysis of fixed computations, not a stronger universal local-lemma theorem.

5 Small state descriptions for local repair

For a finite resampling computation whose selector is determined by the current assignment, that assignment is an exact state; finite controller memory, when present, can be adjoined to it. This state may contain much more information than is needed to compute nontermination. The section separates two notions of sufficient state. A statistic may preserve all weighted counts of surviving paths even when it does not determine the next repair on an individual path. The first property is enough for the survival exponent and the two dimensions studied in this paper; the second is needed to continue the computation exactly.

5.1 Preserving all source powers

Let $\mathcal{X}$ be a finite set of live states. From $x \in \mathcal{X}$, a source outcome e of probability $p(e) > 0$ either moves the computation to a live state $F(x, e)$ or terminates it. For $s \geq 0$, let

$$(B_s)_{xy} = \sum_{e: F(x, e)=y} p(e)^s, \quad x, y \in \mathcal{X}. \quad (39)$$

Thus B_1 is the usual killed transition matrix, whereas the family $(B_s)_{s \geq 0}$ counts source-weighted tape prefixes.

Let $\phi : \mathcal{X} \rightarrow \mathcal{C}$ be a surjection. Say that ϕ *lumps the source powers* if, for every $s \geq 0$ and $c, d \in \mathcal{C}$, the quantity

$$R_s(c, d) := \sum_{e: F(x, e) \in \phi^{-1}(d)} p(e)^s \quad (40)$$

does not depend on the choice of $x \in \phi^{-1}(c)$. This phrase is descriptive rather than standard. At $s = 1$, condition (40) is the usual strong lumpability condition for a Markov chain (Kemeny and Snell, 1960); requiring it for all s is what also preserves the geometry of the source tape.

Proposition 5.1 (Exact source-power reduction). *If ϕ satisfies (40), then, for every $x \in \mathcal{X}$ and integer $T \geq 0$,*

$$(B_s^T \mathbf{1})(x) = (R_s^T \mathbf{1})(\phi(x)). \quad (41)$$

In particular,

$$\rho(B_s) = \rho(R_s). \quad (42)$$

If $R_s r_s = \varrho_s r_s$ for a strictly positive r_s and $\varrho_s > 0$, then $h_s = r_s \circ \phi$ satisfies $B_s h_s = \varrho_s h_s$, and

$$Q_s(e \mid x) = \frac{p(e)^s h_s(F(x, e))}{\varrho_s h_s(x)} \quad (43)$$

for outcomes with $F(x, e) \in \mathcal{X}$ (and zero otherwise) is a probability law on live transitions. Every path $w = e_1 \cdots e_T$ from x_0 to x_T then obeys

$$\log_2 \frac{Q_s[w]}{P[w]} = (1 - s) \log_2 \frac{1}{P[w]} - T \log_2 \varrho_s + \log_2 \frac{h_s(x_T)}{h_s(x_0)}. \quad (44)$$

Consequently, under the hypotheses of Proposition A.1, the survival exponent is $-\log_2 \rho(R_1)$ and the nontermination dimension is the solution of $\rho(R_s) = 1$. The same reduced eigenfunction gives the dimension-attaining Doob transform and the effective-dimension bound.

The proof is given in Appendix E.

The continuum of powers in (40) is not an algorithmic obstacle when the transition system is explicit. Let $\mathcal{R}$ be the finite set of values taken by $p(e)$. Distinct functions r^s , $r \in \mathcal{R}$, are linearly independent as functions of s . Hence (40) is equivalent to requiring, for each $r \in \mathcal{R}$, that the number of probability- r outcomes from x into each class d depend only on $\phi(x)$. When the probabilities are given by exact labels with decidable equality—for example, as rational or algebraic numbers—ordinary labelled partition refinement therefore finds the coarsest source-power lumping in time polynomial in the *explicit* transition graph. This does not by itself avoid an exponentially large assignment graph; it identifies exactly what a smaller representation must preserve.

There is a simple local criterion for Proposition 5.1. Suppose an assignment has an additive statistic $g(x) = \sum_v g_v(x_v)$ and that terminal status depends only on $g(x)$. If, whenever $g(x) = k$, the selected repair block has a fixed old score $c(k)$ and the total s -weight of its redraws with new score j is a number $\nu_{s,k}(j)$ independent of x , then g lumps the source powers, with

$$R_s(k, k - c(k) + j) = \nu_{s,k}(j). \quad (45)$$

This criterion allows the selected block itself to depend on the full assignment. Only its contribution to the statistic, and the source-weighted law of its replacement, must be determined by k .

5.2 Extremal local repair and the Shearer boundary

The next result solves the tape problem for every finite extremal instance of partial rejection sampling. Let $(X_v)_{v \in V}$ be independent finite-valued variables with full-support laws π_v . There is at least one bad event, and A_i depends on the scope $V_i \subseteq V$ and is represented by a set $\mathcal{B}_i \subseteq \prod_{v \in V_i} \mathcal{X}_v$ of bad local configurations. The dependency graph D has vertex set $[m]$ and joins i and j when $V_i \cap V_j \neq \emptyset$. The instance is *extremal* if

$$A_i \cap A_j = \emptyset \quad (ij \in E(D)).$$

Thus two events are either supported on disjoint variables or cannot occur simultaneously. Assume that the satisfying set $\Sigma = \bigcap_i A_i^c$ is nonempty and that every A_i is nonempty. At each step, a fixed deterministic current-assignment rule selects a currently true event and redraws the variables in its scope. Let P be the resulting sequential source law. Assume also that every bad local configuration

has source probability strictly between zero and one. Only the values actually drawn, in their order of revelation, form the tape; unused entries of a resampling table are not part of the tape metric.

For $s \geq 0$, define the source-power mass of event i and of the satisfying frontier by

$$r_i(s) = \sum_{b \in \mathcal{B}_i} \pi_{V_i}(b)^s, \quad Z_\Sigma(s) = \sum_{x \in \Sigma} \pi(x)^s. \quad (46)$$

In particular, $r_i(1) = p_i := \mathbb{P}(A_i)$. If A_i is atomic, meaning that $\mathcal{B}_i$ is a singleton, then $r_i(s) = p_i^s$.

Let

$$P_D(\mathbf{z}) = \sum_{I \subseteq [m] : I \text{ independent in } D} (-1)^{|I|} \prod_{i \in I} z_i \quad (47)$$

be the signed independence polynomial. In the trace monoid $\mathcal{M}(D)$, letters i and j commute exactly when $ij \notin E(D)$. Write $N_i(h)$ for the number of occurrences of i in a trace h , and $|h| = \sum_i N_i(h)$. Set

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{i=1}^m r_i(s)^{N_i(h)}. \quad (48)$$

Theorem 5.2 (Extremal source-power formula). *Let $\mathcal{H}_T$ be the set of minimal consumed tape prefixes ending when the algorithm halts after exactly T repairs. Then, as an identity of formal power series,*

$$\sum_{w \in \mathcal{H}_T} P[w]^s = Z_\Sigma(s) a_T(s), \quad (49)$$

$$\sum_{T \geq 0} u^T \sum_{w \in \mathcal{H}_T} P[w]^s = \frac{Z_\Sigma(s)}{P_D(ur_1(s), \dots, ur_m(s))}. \quad (50)$$

Let $\mathcal{L}_T$ be the set of prefixes for which the computation is still live after T repairs and put $L_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s$. For every fixed $s \geq 0$ there are constants $0 < c_s \leq C_s < \infty$, independent of T , such that

$$c_s a_T(s) \leq L_T(s) \leq C_s a_T(s), \quad T \geq 1. \quad (51)$$

Let $R(s)$ be the radius of convergence in u of the right-hand side of (50). Equivalently, $R(s)$ is the first positive zero of $u \mapsto P_D(u\mathbf{r}(s))$. There is a unique $\delta \in [0, 1]$ such that $R(\delta) = 1$, and

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = R(1)^{-1}, \quad \dim_H^P \mathcal{N} = \delta. \quad (52)$$

Moreover, as a formal power-series identity, and analytically for $|u| < R(1)$,

$$\mathbb{E}u^\tau = \frac{P_D(\mathbf{p})}{P_D(u\mathbf{p})}. \quad (53)$$

The vector $\mathbf{r}(\delta)$ lies on the boundary of Shearer's region for D ; more explicitly,

$$P_D(\mathbf{r}(\delta)) = 0, \quad P_D(u\mathbf{r}(\delta)) > 0 \quad (0 \leq u < 1). \quad (54)$$

If the source, events, and selection rule are computable, the largest effective strong dimension of a nonterminating tape is also δ , and this value is attained.

The proof is given in Appendix F.

The exact source-power masses also give a simple comparison using coarser event data. For a positive vector $\mathbf{z}$, let $\mathcal{R}_D(\mathbf{z})$ denote the first positive zero of $u \mapsto P_D(u\mathbf{z})$.

Corollary 5.3 (Marginal–multiplicity bounds). *Put $k_i = |\mathcal{B}_i|$. Let δ_- and δ_+ be the unique solutions in $[0, 1]$ of*

$$\mathcal{R}_D((p_i^{\delta_-})_i) = 1, \quad \mathcal{R}_D((k_i^{1-\delta_+} p_i^{\delta_+})_i) = 1. \quad (55)$$

Then

$$\delta_- \leq \dim_H^P \mathcal{N} \leq \delta_+. \quad (56)$$

Let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}$$

be the unweighted trace-growth rate. If all $p_i = p$ and all $k_i = k$, this becomes

$$\frac{\log_2 \kappa_D}{\log_2(1/p)} \leq \dim_H^P \mathcal{N} \leq \frac{\log_2(k\kappa_D)}{\log_2(k/p)}. \quad (57)$$

Both bounds coincide when the events are atomic.

The proof is given in Appendix F.

For atomic events, Theorem 5.2 takes a particularly simple form.

Corollary 5.4 (Powered-marginal Shearer boundary). *If every bad event is atomic, then*

$$\dim_H^P \mathcal{N} = \delta, \quad \mathbf{p}^\delta = (p_1^\delta, \dots, p_m^\delta) \text{ lies on the Shearer boundary of } D. \quad (58)$$

The same δ is the largest effective strong dimension under computability. If, in addition, $p_i = p$ for every i , put

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

Then

$$\mathbb{P}\{\tau \geq T\}^{1/T} \rightarrow p\kappa_D, \quad \delta = \frac{\log_2 \kappa_D}{\log_2(1/p)} = 1 - \frac{\alpha}{\log_2(1/p)}. \quad (59)$$

For unequal atomic probabilities, with $\ell_{\min} = \min_i \log_2(1/p_i)$ and $\ell_{\max} = \max_i \log_2(1/p_i)$,

$$1 - \frac{\alpha}{\ell_{\min}} \leq \delta \leq 1 - \frac{\alpha}{\ell_{\max}}. \quad (60)$$

The proof is given in Appendix F.

Relation to classical trace enumeration. The confluence of extremal partial rejection sampling, the correspondence between transcripts and Cartier–Foata traces, and the identity $1/P_D$ are classical (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). Equation (53) is already implicit in that analysis. The additional step here is to power every consumed source value before summing. This identifies the whole source-power pressure, and hence the Hausdorff and pointwise dimensions of nontermination, with the point at which $\mathbf{r}(s)$ reaches the same boundary. For non-atomic events it is important to use $r_i(s)$ from (46); replacing it by p_i^s is generally incorrect (indeed $r_i(s) \geq p_i^s$ for $0 \leq s \leq 1$).

5.3 Repairing disagreements on a graph

Let $G = (V, E)$ be any connected simple graph with $N = |V| \geq 2$. Draw independent $X_v \sim \text{Bernoulli}(p)$, where $0 < p < 1$, and put $q = 1 - p$. For each edge $e = \{u, v\}$, define

$$A_e = \{X_u \neq X_v\}.$$

Fix any deterministic nonanticipating rule that selects a true event; the rule may use the current assignment and the past repair history, but not future source outcomes. To repair A_e , redraw X_u and X_v independently from their source laws. The terminal assignments are 0^V and 1^V . When G has two adjacent edges, the repairs genuinely overlap. For two distinct edges sharing one endpoint,

$$\mathbb{P}(A_e) = 2pq, \quad \mathbb{P}(A_e \cap A_f) = pq. \quad (61)$$

Let τ be the number of repairs before a constant assignment is reached, with $\tau = 0$ if the initial assignment is constant. Let P be the sequential source law of the initial assignment and all redraws, let $\mathcal{N}(G, p) = \{\tau = \infty\}$, and for a redraw word $w = z_1 \cdots z_T$ write $P_p[w] = \prod_{t=1}^T P_p(z_t)$, where $P_p(z) = p^{z_1+z_2} q^{2-z_1-z_2}$ for $z = (z_1, z_2)$. For $s \geq 0$, define the $(N-1) \times (N-1)$ tridiagonal matrix R_s on $\{1, \dots, N-1\}$ by

$$R_s(k, k-1) = q^{2s}, \quad R_s(k, k) = 2(pq)^s, \quad R_s(k, k+1) = p^{2s}, \quad (62)$$

where entries outside $\{1, \dots, N-1\}$ are omitted.

Theorem 5.5 (Exact nontermination for disagreement repair). *For every connected G , every deterministic nonanticipating selection rule, and every integer $T \geq 0$, the entire law of τ is the same. More precisely,*

$$\mathbb{P}\{\tau > T\} = \sum_{k=1}^{N-1} \binom{N}{k} p^k q^{N-k} (R_1^T \mathbf{1})_k. \quad (63)$$

More generally, the source-power sum of all tapes whose initial assignment and next T redraws remain live is

$$Z_T(s) = \sum_{k=1}^{N-1} \binom{N}{k} p^{sk} q^{s(N-k)} (R_s^T \mathbf{1})_k. \quad (64)$$

Set

$$\vartheta = \frac{\pi}{N}, \quad c_N = \cos\left(\frac{\pi}{2N}\right).$$

The exact source-power rate, survival rate, and source-relative dimension are

$$\varrho_s = \lim_{T \rightarrow \infty} Z_T(s)^{1/T} = \rho(R_s) = 4(pq)^s c_N^2, \quad (65)$$

$$\lambda_{N,p} = \lim_{T \rightarrow \infty} \mathbb{P}\{\tau > T\}^{1/T} = 4pq c_N^2, \quad (66)$$

$$\dim_H^P \mathcal{N}(G, p) = \delta_{N,p} = \frac{\log(4c_N^2)}{\log(1/(pq))}. \quad (67)$$

For computable data, including a computable selection rule, $\delta_{N,p}$ is also the largest effective strong dimension of a nonterminating tape, and it is attained.

The dimension-attaining Doob transform is explicit. Start from any fixed computable live assignment; this finite initial choice does not affect dimension. If the current assignment has k ones and a redraw $z \in \{00, 01, 10, 11\}$ has $j(z)$ ones, put $d(z) = j(z) - 1$ and

$$Q^*(z | k) = \frac{\sin((k + d(z))\vartheta)}{4c_N^2 \sin(k\vartheta)}, \quad (68)$$

with value zero when $k + d(z) \in \{0, N\}$. This conditional law for the four redraw outcomes is independent of p , the graph, and the selection rule. Its Hamming-weight marginal is reversible with stationary distribution

$$\nu^*(k) = \frac{2}{N} \sin^2(k\vartheta), \quad 1 \leq k < N. \quad (69)$$

For every $s \geq 0$ and every surviving path $w = z_1 \cdots z_T$ from k_0 to k_T ,

$$\log_2 \frac{Q^*[w \mid k_0]}{P_p[w]} = (1-s)I_p(w) - T \log_2 \varrho_s + \log_2 \frac{h_s(k_T)}{h_s(k_0)}, \quad (70)$$

where

$$h_s(k) = \left(\frac{q}{p}\right)^{sk} \sin(k\vartheta), \quad I_p(w) = \log_2 \frac{1}{P_p[w]}. \quad (71)$$

Set $h_s(0) = h_s(N) = 0$ when a boundary value occurs in a recurrence. Thus the same change of measure gives the survival exponent at $s = 1$ and the dimension bound at $s = \delta_{N,p}$.

The proof is given in Appendix G.

There is a particularly simple information identity behind the theorem. For one redraw,

$$P_p(z) = pq \left(\frac{p}{q}\right)^{d(z)}.$$

Consequently every surviving path satisfies

$$I_p(w) = T \log_2 \frac{1}{pq} + (k_T - k_0) \log_2 \frac{q}{p}. \quad (72)$$

The second term is bounded independently of T . If $\alpha_{N,p} = -\log_2 \lambda_{N,p}$, then

$$\boxed{\delta_{N,p} = 1 - \frac{\alpha_{N,p}}{\log_2(1/(pq))}.} \quad (73)$$

Equivalently, if $c = \log_2(1/(pq))$ is the source information per redraw up to the bounded endpoint term and $h = \log_2(4c_N^2)$ is the surviving information rate, then $\alpha_{N,p} = c - h$ and $\delta_{N,p} = h/c$. Thus $\alpha_{N,p} = c(1 - \delta_{N,p})$.

For a fair source,

$$\lambda_{N,1/2} = c_N^2, \quad \delta_{N,1/2} = 1 + \log_2 c_N, \quad \delta_{N,1/2} = 1 - \frac{\alpha_{N,1/2}}{2}. \quad (74)$$

Moreover,

$$\alpha_{N,1/2} = \frac{\pi^2}{4N^2 \ln 2} + O(N^{-4}), \quad 1 - \delta_{N,1/2} = \frac{\pi^2}{8N^2 \ln 2} + O(N^{-4}). \quad (75)$$

The mean absorption time is also exact:

$$\mathbb{E}[\tau \mid M_0 = k] = 2k(N - k), \quad \mathbb{E}\tau = \frac{N(N - 1)}{2}. \quad (76)$$

Thus the process terminates almost surely and has quadratic mean running time, while the dimension of its nonterminating tapes tends to one.

The same result has a standard constraint-satisfaction form. Let $G = (L \cup R, E)$ be connected and bipartite, draw $X_v \sim \text{Bernoulli}(p)$ on L and $X_v \sim \text{Bernoulli}(q)$ on R , and declare a monochromatic

edge bad. Repairing a bad edge redraws its endpoints from their respective source laws. The change of variables

$$Y_v = X_v \quad (v \in L), \quad Y_v = 1 - X_v \quad (v \in R)$$

turns this process into disagreement repair with i.i.d. Bernoulli(p) variables. Therefore Theorem 5.5 gives the exact time and tape geometry of local resampling for a proper two-colouring, for every connected bipartite graph and every deterministic repair rule.

5.4 A small counting state versus a large continuation state

The statistic M determines all the quantities in Theorem 5.5, but it does not determine the next edge selected by a specified rule. This distinction cannot be removed merely by assuming that the event–variable incidence graph has small pathwidth.

Proposition 5.6 (A pathwidth-one state-cardinality obstruction). *Let G be the path on vertices $0, 1, \dots, N-1$, and always repair the leftmost disagreeing edge. Suppose a self-contained deterministic state summary has two properties: its current value determines the next repair (or halt), and it updates exactly from that value and the realized redraw. Then it has at least 2^{N-1} states when the halting state is included, and at least $2^{N-1} - 1$ live states.*

The proof is given in Appendix G.

The incidence graph in Proposition 5.6 is itself a path and has pathwidth one. Hence bounded pathwidth alone does not imply a $2^{O(w)}$ poly(N) exact continuation state. This is a state-cardinality, or continuation-equivalence, lower bound: it does not assert that the scalar survival exponent is computationally hard. Indeed, the same example has the exact $(N-1)$ -state pressure reduction above. Thus a statistic sufficient for pressure need not suffice to continue a run exactly.

Relation to earlier processes. The reduction of a two-colour disagreement process to the number of vertices of one colour is closely related to gambler’s-ruin reductions for discordant voting (Cooper et al., 2018). In that model a selected discordant edge is resolved by copying one endpoint, whereas here both endpoints are independently redrawn. The novelty claimed here is not the birth–death calculation by itself, but its simultaneous extension to every source-power operator, the likelihood identity (70), the tape-dimension conclusions, and the separation between an exact pressure state and an exact continuation state. The instance is also nonextremal: adjacent bad events can occur together. Thus the extremal-instance analysis of partial rejection sampling does not yield this reduction (Guo et al., 2019; Jerrum, 2024).

5.5 Adjacent rises: an extremal benchmark

Let $n \geq 1$ and let $X_0, \dots, X_n$ be independent bits with $\mathbb{P}\{X_i = 1\} = p_i \in (0, 1)$, and set

$$A_i = \{X_{i-1} = 0, X_i = 1\}, \quad 1 \leq i \leq n.$$

At each step, any deterministic current-assignment rule may select a true A_i ; the repair redraws (X_{i-1}, X_i) from its product marginal. For $\theta \geq 0$, let P be the sequential source law and let B_θ be the full live-state source-power operator. For a pair outcome $z = (z_{i-1}, z_i)$, write

$$P_i(z) = (1 - p_{i-1})^{1-z_{i-1}} p_{i-1}^{z_{i-1}} (1 - p_i)^{1-z_i} p_i^{z_i}.$$

Define the symmetric tridiagonal matrix K_θ , indexed by $0, \dots, n$, by

$$(K_\theta)_{i-1,i} = (K_\theta)_{i,i-1} = [(1 - p_{i-1})p_i]^{\theta/2}, \quad (K_\theta)_{ii} = 0. \quad (77)$$

For the state lower bound below, an *outcome-driven state representation* is a map σ from assignments to a finite set together with a status map and an update map U such that the state alone reports live versus halted and

$$\sigma(F(x, z)) = U(\sigma(x), z)$$

whenever the deterministic rule repairs the live assignment x and observes the pair outcome $z \in \{00, 01, 10, 11\}$; here $F(x, z)$ is the resulting assignment. Thus no part of the current assignment is available outside the state.

Theorem 5.7 (Adjacent-rise pressure). *For every deterministic current-assignment selection rule and every $\theta \geq 0$,*

$$\rho(B_\theta) = \rho(K_\theta)^2. \quad (78)$$

Consequently,

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \rho(K_1)^2, \quad \dim_H^P \mathcal{N} = \delta, \quad \rho(K_\delta) = 1. \quad (79)$$

The root $\delta \in [0, 1]$ is unique. If the biases and selection rule are computable, it is also the largest effective strong dimension of a nonterminating tape and is attained by an explicit edgewise product-form law.

For the least-index rule, however, every outcome-driven state representation has at least 2^n states, including the halting state.

The proof is given in Appendix H.

Corollary 5.8 (A linear eigenproblem versus exponential continuation state). *For fixed θ , the matrix K_θ has $2n$ nonzero entries and is constructed in linear time. Its largest eigenvalue, and hence $\rho(B_\theta)$, can be approximated to additive error ε using $O(n \log(1/\varepsilon))$ arithmetic operations by Sturm bisection for a symmetric tridiagonal matrix. The dimension δ is then obtained by one-dimensional bisection in θ . This is an arithmetic-operation statement; bit complexity also depends on how the biases are represented and on numerical conditioning. In contrast, the least-index process has 2^n outcome-driven continuation states.*

The proof is given in Appendix H.

Adjacent events A_i and A_{i+1} are mutually exclusive, while nonadjacent events are independent. Thus this is an extremal instance in the terminology of partial rejection sampling: dependent bad events cannot occur simultaneously (Guo et al., 2019; Jerrum, 2024). In this regime, known Cartier–Foata enumeration identifies repair transcripts with traces and gives the generating function $P_D(\mathbb{P}(A_1)z_1, \dots, \mathbb{P}(A_n)z_n)^{-1}$, where P_D is the signed independence polynomial of the dependency graph. The theorem above uses the same structure at every source power and makes its tape-geometric consequence explicit; it does not claim a new conditional-output theorem or a new trace enumeration.

For the dependency path D , put

$$q_\theta = P_D(\mathbb{P}(A_1)^\theta, \dots, \mathbb{P}(A_n)^\theta) = \sum_{I \in \mathcal{I}(D)} (-1)^{|I|} \prod_{i \in I} \mathbb{P}(A_i)^\theta.$$

The tridiagonal determinant recurrence gives $q_\theta = \det(I - K_\theta)$. At $\theta = \delta$ the full determinant vanishes; deleting any event strictly lowers the Perron root, so every proper subfamily still has positive Shearer polynomial. Hence the powered marginal vector reaches the Shearer boundary exactly at $\theta = \delta$ (Shearer, 1985; Scott and Sokal, 2005). The determinant and boundary description are classical. The conclusion here is that this boundary value is also the source-relative dimension of the nonterminating tapes of the sequential resampling computation.

6 Conclusion

A random tape turns a randomized algorithm into a deterministic computation, so nontermination can be studied both as a rare event and as a set of individual sequences. The backward identity makes this connection exact at finite time: the same likelihood ratio bounds the probability of continued execution and the description length of every tape prefix that produces it. The Rényi power sums of live cylinders then pass from this pointwise comparison to the Hausdorff dimension of the whole exceptional set.

The paper computes these quantities in two local-repair regimes. For extremal partial rejection, powering the Cartier–Foata trace weights places the nontermination dimension at the Shearer boundary. For nonextremal disagreement repair, Hamming weight preserves every source power and gives an exact birth–death solution independent of the graph’s topology and the nonanticipating repair rule. This yields the full stopping law and Hausdorff dimension, together with an explicit dimension-attaining change of measure; under computability it also gives the maximal effective strong dimension.

The resulting computational question concerns state size. A dependency graph may discard intersection data needed for exact answers, whereas the full assignment, together with any finite controller memory, is exact but generally exponential. The path examples show that a small state can preserve survival and dimension even when exact continuation requires exponentially many states. In the language of localization, the live cylinders are the region retained by the computation; the challenge is to summarize that region without losing its probability or geometry. This paper does not yet provide such a summary for arbitrary overlapping repairs.

A Finite-state computations

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0 . Each outgoing edge $e : s \rightarrow t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \dots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0 . Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (80)$$

For any square matrix C , its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ -weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

Use $P[w]$ as the diameter of the cylinder $[w]$. The resulting Hausdorff dimension is the source-relative dimension $\dim_H^P$ defined in the introduction; for a uniform q -ary source it is ordinary Hausdorff dimension in the q -adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\text{eff}, P}$ by replacing the limsup with a liminf.

Proposition A.1 (The finite-state formula). *There is a unique $\delta \in [0, 1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (81)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (82)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (83)$$

If $B_\delta h = h$ with $h_s > 0$ for every state s , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (84)$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^n Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (85)$$

For a computable system with computable Q_δ , every nonterminating tape satisfies $\text{Dim}_{\text{eff},P}(\omega) \leq \delta$. A tape that is Martin–Löf random under Q_δ satisfies $\dim_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$.

Proof. Since the graph contains a directed cycle, $\rho(B_0) \geq 1$. Since B_1 is substochastic, $\rho(B_1) \leq 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ , because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ . The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (81); setting $\theta = 1$ gives (82).

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ -weight of the length- n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\dim_H^P \mathcal{N} \leq \delta$. At $\theta = \delta$, (84) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w , such that $Q_\delta[w] \leq CP[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\dim_H^P \mathcal{N} \geq \delta$. It therefore gives the required lower bound, proving (83) and (85) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system, coding under Q_δ gives, for every surviving prefix w of length n ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ -Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). $\square$

Under a uniform q -ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t . Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\dim_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\text{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \quad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (85) is the finite-state form of the comparison used in Theorem 3.1.

B Proof for rejection sampling

Proof of Proposition 2.1. For $|U| \geq 2$, the root lies in $(0, 1)$. Since $\pi(x) = p\nu(x)$ on U , the defining equation for δ becomes

$$1 = p^\delta \sum_{x \in U} \nu(x)^\delta.$$

Taking logarithms gives $\delta\alpha = (1 - \delta)H_\delta(\nu)$, which proves (5). The bounds follow from $H(\nu) \leq H_\delta(\nu) \leq \log_2 |U|$. $\square$

C Proofs for the backward likelihood identity

Proof of Theorem 3.1. Take the ratio of (8) and (7). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (9).

It remains to use this likelihood ratio. Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce at least T repairs, and let $\mathcal{R}_T$ be the set of formal records on which G_T is defined. For $w \in \mathcal{W}_T$, write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that R_T is injective.

First observe that G_T has total mass at most one. Indeed, for a fixed list $\ell = (A_1, \dots, A_T)$, the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of R_T gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus μ_T is a subprobability mass function. Its total mass can be strictly less than one because Q_T may itself have mass less than one, or because G_T may assign mass to formal records that are not produced by the algorithm.

For every $w \in \mathcal{W}_T$, the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because $\Delta_T = -\infty$ outside E_T and the cylinders determined by $\mathcal{W}_T$ are disjoint, for every $u \geq 0$,

$$\begin{aligned}
\mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\
&= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\
&\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\
&\leq 2^{-u}.
\end{aligned}$$

This proves (10); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Finally, uniform computability of the instance and of $(Q_T)_{T \geq 1}$ makes $(\mu_T)_{T \geq 1}$ a uniformly computable family of subprobability mass functions. Shannon–Fano coding, or equivalently the coding theorem for computable subprobability masses (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019), therefore gives

$$K(\Xi_T \mid T, \text{instance}) \leq -\log_2 \mu_T(\Xi_T) + O(1).$$

The constant is uniform in T and in the realized run. Since

$$-\log_2 \mu_T(\Xi_T) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T,$$

rearranging proves (11). $\square$

Proof of Corollary 3.2. On $\mathcal{W}_T$,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking P_T -expectations gives (12). For the second claim, $\Delta_T = S_T + \log_2 Q_T(L_T)$. If r_T is the conditional law of L_T given E_T , then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \parallel Q_T)$$

when Q_T has total mass one; allocating less mass can only decrease the left side. The maximum is attained at $Q_T = r_T$. $\square$

Proof of Corollary 3.4. The assumptions give $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$. Equation (17) follows from (10). Equation (11), together with the $O(\log_2 T)$ bits needed to specify T , gives $K(\Xi_T) \leq c_0 + cT - (\beta - \gamma)T + o(T)$ along an infinite execution. If $\beta - \gamma > c$, this is impossible for all large T , so no infinite run exists. Otherwise divide by $c_0 + cT$; the gap between consecutive repair prefixes is the fixed number c of bits. $\square$

D Proof for whole-block resampling

Proof of Theorem 4.1. The function W is continuous and strictly decreasing because $0 < \pi(x) < 1$. Moreover, $W(0) = |U| \geq 1$ and $W(1) = \pi(U) \leq 1$, so (29) has a unique solution in $[0, 1]$.

For full nonterminal states $x, y \in U$, resampling the common scope gives $B_{\theta}(x, y) = \pi(y)^{\theta}$. Hence, for every $x \in C_A$,

$$\sum_{y \in C_B} B_{\theta}(x, y) = \sum_{y \in C_B} \pi(y)^{\theta} = w_B(\theta),$$

independently of x . Thus grouping states by the sets C_A leaves the weighted transition calculation unchanged and proves (33). The reduced matrix is $\mathbf{1} w(\theta)^\top$, so its only nonzero eigenvalue is $W(\theta)$ and $\mathbf{1}$ is a right eigenvector. Proposition A.1 now gives $\dim_H^P \mathcal{N} = \delta$. Each repair draws a fresh block from π and continues precisely on U , proving the survival formula and its uniform specialization.

For the dimension bound, observe that Q_δ is computable and supported on U . Every $\omega \in U^\mathbb{N}$ therefore satisfies

$$K(\omega_{1:n}) \leq -\log_2 Q_\delta^\mathbb{N}[\omega_{1:n}] + O(\log_2 n) = \delta(-\log_2 P[\omega_{1:n}]) + O(\log_2 n).$$

Division by the source information proves (30). If ω is Martin–Löf random for $Q_\delta^\mathbb{N}$, Levin–Schnorr gives the reverse inequality up to $O(1)$, and hence equality. This also proves (4).

It remains to construct the record laws. The repair list is determined by the overwritten states because $A_t = \phi(Y_t)$. Reversal recovers the used tape from $(S_T, Y_{1:T})$. The source factorization is

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T \pi(Y_t).$$

Since (29) holds, the record law

$$G_{\delta,T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^T \left[\pi(y_t)^\delta \mathbf{1}\{y_t \in U\} \right]$$

has total mass one. Dividing by the source factorization proves (35). Similarly, define

$$G_{\text{surv},T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^T \frac{\pi(y_t) \mathbf{1}\{y_t \in U\}}{\pi(U)}.$$

This is a probability law, and division by the source factorization proves (36). The region containing y_t determines the recorded rejection label $A_t = \phi(y_t)$. $\square$

E Proof for source-power reduction

Proof of Proposition 5.1. Let C lift a function on $\mathcal{C}$ to a function on $\mathcal{X}$ by $(Cf)(x) = f(\phi(x))$. Condition (40) is exactly $B_s C = C R_s$. Since $C\mathbf{1} = \mathbf{1}$, iteration gives (41). For a nonnegative matrix M , $\|M\|_\infty = \max_x (M\mathbf{1})(x)$. Surjectivity of ϕ and (41) therefore give

$$\|B_s^T\|_\infty = \|R_s^T\|_\infty \quad \text{for every } T.$$

Taking T th roots proves (42) by Gelfand’s formula. The eigenfunction statement follows again from $B_s C = C R_s$. Equation (43) has row sum one, and multiplication along a path telescopes to

$$Q_s[w] = P[w]^s \varrho_s^{-T} \frac{h_s(x_T)}{h_s(x_0)},$$

which is (44). $\square$

F Proofs for extremal local repair

Proof of Theorem 5.2. The classical Cartier–Foata identity gives

$$\sum_{h \in \mathcal{M}(D)} \prod_i z_i^{N_i(h)} = \frac{1}{P_D(\mathbf{z})} \quad (86)$$

as a formal power series (Cartier and Foata, 1969; Jerrum, 2024). For extremal partial rejection sampling, traces are in bijection with resampling transcripts: swapping two consecutive nonadjacent labels swaps repairs on disjoint scopes and changes neither the resampling table nor its frontier. The diamond property also makes the transcript independent of the rule used to choose among currently true events (Guo et al., 2019; Jerrum, 2024).

Fix a trace h . Its resampling blocks and final frontier partition the entries of the consumed resampling table. A halting table with transcript h is obtained by putting a configuration from $\mathcal{B}_i$ in every block labelled i and a satisfying assignment in the final frontier. Conversely, every such choice realizes the transcript. One way to see the converse is to use the Cartier–Foata layers of h and read them backward. Writing a bad configuration into every scope of one layer makes exactly the events of that layer true: their scopes are disjoint; their neighbors are excluded by extremality; and every nonneighbor is unchanged from the next layer. The Cartier–Foata condition says that every event in the next layer is a neighbor of, or equal to, an event in the current layer. Induction from the satisfying frontier proves the claim.

All cells in this partition are independent under the source. After taking the s th power and summing their values, a block labelled i contributes $r_i(s)$ and the final frontier contributes $Z_\Sigma(s)$. Thus a trace h contributes

$$Z_\Sigma(s) \prod_i r_i(s)^{N_i(h)}.$$

Summing first over traces of length T and then over T , and applying (86), proves (49)–(50).

The same partition proves the upper bound in (51), with $C_s = \sum_x \pi(x)^s$: simply allow every full assignment at the current frontier. The lower bound must also respect the fixed sequential selector. Fix a satisfying assignment y and, for each i , a configuration $b_i \in \mathcal{B}_i$. Take any trace of length T and any choice of bad configurations in its blocks, put y in the final frontier, and run the fixed selector. This is a halting tape with that trace. Let i be the event repaired last. Every cell of the final frontier on V_i is drawn by this last repair. Replace just those newly drawn values, from $y|_{V_i}$ to b_i . No earlier state or repair choice changes, while the state after the T th repair has exactly A_i true: its neighbors are excluded by extremality and its nonneighbors have disjoint scopes and remain false.

This map from the tables with fixed final frontier y to live prefixes is injective: the run identifies its last label i , after which restoring $y|_{V_i}$ recovers the original table. It changes source probability by the factor $\pi_{V_i}(b_i)/\pi_{V_i}(y|_{V_i})$. Consequently the lower bound holds with

$$c_s = \pi(y)^s \min_i \left(\frac{\pi_{V_i}(b_i)}{\pi_{V_i}(y|_{V_i})} \right)^s > 0.$$

This proves (51) without assuming that the selector finishes one Cartier–Foata layer before starting the next.

By Cauchy–Hadamard and (51), the source-power growth of live prefixes is $R(s)^{-1}$. At $s = 1$, extremality and inclusion–exclusion give $Z_\Sigma(1) = \mathbb{P}(\Sigma) = P_D(\mathbf{p}) > 0$. More is true. For any $\mathbf{y}$ with $0 \leq y_i \leq p_i$, independently retain event A_i with probability y_i/p_i . The retained events are still

extremal, their marginals are $\mathbf{y}$, and their common avoidance event contains Σ . Inclusion–exclusion therefore gives

$$P_D(\mathbf{y}) \geq \mathbb{P}(\Sigma) > 0 \quad (0 \leq \mathbf{y} \leq \mathbf{p}).$$

Compactness and continuity then give $P_D(u\mathbf{p}) > 0$ for all $u \leq 1 + \varepsilon$, for some $\varepsilon > 0$. Thus $R(1) > 1$. Summing the exact halting probabilities at $s = 1$ gives (53). The first assertion of (52) follows; the existence of the exponential-rate limit follows from the standard finite Cartier–Foata automaton.

Every entry of every bad local configuration has source probability strictly between zero and one. Put $c = \max_{i,b \in \mathcal{B}_i} \pi_{V_i}(b) < 1$. If $t > s$, then $r_i(t) \leq c^{t-s} r_i(s)$ for every i , and hence $a_T(t) \leq c^{(t-s)T} a_T(s)$. The finite Cartier–Foata automaton also shows that the growth $R(s)^{-1}$ is continuous. It is therefore strictly decreasing, is less than one at $s = 1$, and is at least one at $s = 0$ because the trace i^T exists for every T . This proves the existence and uniqueness of δ .

For completeness, the same conclusion can be read directly from the full assignment chain for the fixed selection rule. Its total s -weight after T live repairs is $L_T(s)$; the initial assignment contributes only a fixed finite starting vector. Decompose its finite nonterminal graph into reachable cyclic strongly connected components and apply Proposition A.1 to each component. Equation (51) says that the largest component pressure is $\log R(s)^{-1}$, so the largest component root is δ . This proves $\dim_H^P \mathcal{N} = \delta$. Every infinite path eventually remains in one of these finitely many components. Under computability, coding in that component gives the upper effective-strong-dimension bound, and a Martin–Löf-random path for a critical component attains δ .

Finally, the trace series has nonnegative coefficients. Pringsheim’s theorem therefore implies that its first singularity on every positive ray occurs at a positive zero of P_D . At $s = \delta$ that singularity is $u = 1$, which proves (54). Equivalently, the open domain of convergence of (86) consists of the vectors $\mathbf{z}$ for which $P_D(\mathbf{y}) > 0$ for every $0 \leq \mathbf{y} \leq \mathbf{z}$; this is the standard independence-polynomial description of Shearer’s region (Shearer, 1985; Scott and Sokal, 2005). $\square$

Proof of Corollary 5.3. For $0 \leq s \leq 1$, concavity gives, event by event,

$$p_i^s = \left(\sum_{b \in \mathcal{B}_i} \pi_{V_i}(b) \right)^s \leq r_i(s) \leq k_i^{1-s} p_i^s.$$

The trace coefficients in (48) are coordinatewise increasing in these weights. Their exponential growth rates are strictly decreasing in s , so comparison at every s brackets their critical roots and proves (56). When the weights are constant across vertices, the trace growth is κ_D times that common weight. Solving the two scalar equations gives (57). $\square$

Proof of Corollary 5.4. Atomicity gives $r_i(s) = p_i^s$. In the equal-probability case, $a_T(s) = p^{sT} a_T(0)$, so its exponential growth is $p^s \kappa_D$. Setting $s = 1$ gives the survival rate and setting $s = \delta$ makes the growth equal to one, proving (59). In general, write $\alpha = -\log_2 \gamma(1)$, where $\gamma(s)$ is the trace growth at weights p_i^s . For $0 \leq s \leq 1$, each length- T trace satisfies

$$2^{(1-s)\ell_{\min} T} \prod_i p_i^{N_i(h)} \leq \prod_i p_i^{sN_i(h)} \leq 2^{(1-s)\ell_{\max} T} \prod_i p_i^{N_i(h)}.$$

Taking exponential growth rates and using $\gamma(\delta) = 1$ proves (60). $\square$

G Proofs for disagreement repair

Proof of Theorem 5.5. Let $M(x) = \sum_{v \in V} x_v$. Every selected edge contains one zero and one one. If its redraw contains J ones, then

$$M_{t+1} = M_t - 1 + J, \quad J \sim \text{Binomial}(2, p). \quad (87)$$

Because G is connected, an assignment is live exactly when $1 \leq M \leq N - 1$. Conditional on any past with $M_t = k$, the four possible redraws have s -weights $q^{2s}, (pq)^s, (pq)^s, p^{2s}$ and move the count by $-1, 0, 0, +1$, respectively. Induction on T therefore shows directly that the source-power sums grouped by M evolve by (62), even when the selector uses the past. The initial value is $\text{Binomial}(N, p)$, proving (63) and (64). This also proves that the full absorption-time law is independent of G and of the selection rule.

For $1 \leq k < N$, the positive function h_s in (71) satisfies

$$\begin{aligned} (R_s h_s)(k) &= q^{2s} h_s(k-1) + 2(pq)^s h_s(k) + p^{2s} h_s(k+1) \\ &= 4(pq)^s c_N^2 h_s(k). \end{aligned}$$

Proposition 5.1 therefore gives (65). Equivalently, R_s is diagonally similar to $(pq)^s$ times the tridiagonal matrix with diagonal 2 and off-diagonal 1; its eigenvalues are

$$4(pq)^s \cos^2\left(\frac{r\pi}{2N}\right), \quad r = 1, \dots, N-1. \quad (88)$$

Equations (66) and (67) follow by setting $s = 1$ and by solving $\varrho_s = 1$, respectively.

Applying (43) with h_s gives

$$\frac{P_p(z)^s h_s(k + d(z))}{\varrho_s h_s(k)} = \frac{\sin((k + d(z))\vartheta)}{4c_N^2 \sin(k\vartheta)}.$$

The right-hand side is independent of s and p , and its sum over the four redraws is one because

$$\sin((k-1)\vartheta) + 2\sin(k\vartheta) + \sin((k+1)\vartheta) = 4c_N^2 \sin(k\vartheta).$$

Moreover, $\nu^*(k)Q^*(11 \mid k) = \nu^*(k+1)Q^*(00 \mid k+1)$; the self-loops need no check. This proves reversibility and (69). (70) is (44). At the root $s = \delta_{N,p}$, endpoint ratios are bounded above and below because $1 \leq k < N$. Equation (64) gives the Hausdorff upper bound directly. The global positive law Q^* and the bounded likelihood ratio give the matching mass-distribution lower bound and, under computability, the effective-dimension upper bound and attainment on a Q^* -Martin-Löf-random tape. This argument does not require the full assignment chain for a particular selector to be irreducible. $\square$

Proof of Proposition 5.6. For each $S \subseteq \{1, \dots, N-1\}$, let $x_0^S = 0$ and let $x_i^S = 1$ exactly when $i \in S$. Give every repair the common future redraw 00. The successive repair labels are the elements of S in increasing order, after which the computation halts. Distinct sets therefore have distinct output sequences under the same continuation. If two of the x^S had the same summary, deterministic prediction and updating would give the same output sequence, a contradiction. $\square$

H Proofs for adjacent rises

Proof of Theorem 5.7. For a live word $x = x_0 \cdots x_n$, let $a(x)$ be its first zero and $b(x)$ its last one. Then $a(x) < b(x)$, and the sets

$$C_{a,b} = \{1^a 0 u 10^{n-b} : u \in \{0, 1\}^{b-a-1}\}, \quad 0 \leq a < b \leq n,$$

partition the live assignments. Every repair can only increase a or decrease b . Hence B_θ is block upper triangular over the intervals $[a, b]$.

Fix one diagonal block and write

$$\alpha_j = (1 - p_j)^\theta, \quad \beta_j = p_j^\theta, \quad c_i = \sqrt{\alpha_{i-1} \beta_i}.$$

Let η and $v = (v_a, \dots, v_b) > 0$ be the Perron value and vector of the weighted path $K_{\theta, [a, b]}$. For $a < j < b$, put

$$t_j = \sqrt{\frac{\alpha_{j-1} \alpha_j}{\beta_j \beta_{j+1}}} \frac{v_{j-1}}{v_{j+1}}, \quad h(x) = \prod_{j=a+1}^{b-1} t_j^{x_j}. \quad (89)$$

Set $t_a = 0$ and $t_b = +\infty$ for the boundary formulas. The Perron recurrence gives, for $a \leq j < b$,

$$U_j := \alpha_j + \beta_j t_j = \eta \sqrt{\frac{\alpha_j}{\beta_{j+1}}} \frac{v_j}{v_{j+1}}, \quad t_i = \frac{U_{i-1} U_i}{\eta^2} \quad (a < i < b). \quad (90)$$

If the selected rise is i , summing $P_i(z)^\theta h(y)/h(x)$ over the outcomes that retain the endpoints of $[a, b]$ gives

$$\frac{(B_{\theta, [a, b]} h)(x)}{h(x)} = \begin{cases} \alpha_a(\alpha_{a+1}/t_{a+1} + \beta_{a+1}), & i = a + 1, \\ U_{i-1}(\alpha_i/t_i + \beta_i), & a + 1 < i < b, \\ \beta_b U_{b-1}, & i = b. \end{cases}$$

Equations (90) and the two endpoint Perron equations make each expression equal to η^2 . The same is immediate when $b = a + 1$. Thus $B_{\theta, [a, b]} h = \eta^2 h$, independently of which true rise the rule selects. Since $h > 0$, Collatz–Wielandt gives

$$\rho(B_{\theta, [a, b]}) = \rho(K_{\theta, [a, b]})^2.$$

Every proper interval matrix is a proper principal submatrix of the irreducible K_θ , so strict Perron comparison makes $[0, n]$ the unique maximal block. This proves (78); the finite-state pressure formula gives (79). For $n \geq 2$, $\rho(K_0) = 2 \cos(\pi/(n+2)) > 1$. On the other hand, from every live word a sequence of at most $n+1$ outcomes 00 removes the right endpoint of one selected rise at each step and reaches a terminal nonincreasing word $1^a 0^{n+1-a}$. The probability of this sequence is bounded below uniformly over the finite state space. Hence $\rho(B_1) < 1$ and therefore $\rho(K_1) < 1$. Strict monotonicity of $\rho(K_\theta)$ now gives a unique root in $(0, 1)$. For $n = 1$, $\rho(K_0) = 1$ and $\delta = 0$.

The law attaining the dimension is also local. At $\theta = \delta$, use the preceding t_j on $C_{0, n}$ and define

$$U_j = \alpha_j + \beta_j t_j, \quad V_j = \alpha_j/t_j + \beta_j, \quad U_{i-1} V_i = 1.$$

Here $t_0 = 0$ and $t_n = +\infty$, interpreted by limits. Thus U_j is used for $0 \leq j < n$ and V_j for $0 < j \leq n$. If rise i is selected, redraw its two coordinates independently according to

$$\begin{aligned} L_j(0) &= \alpha_j/U_j, & L_j(1) &= \beta_j t_j/U_j, \\ R_j(0) &= \alpha_j/(t_j V_j), & R_j(1) &= \beta_j/V_j, \end{aligned}$$

using $L_{i-1} \otimes R_i$. The boundary marginals are degenerate, so the run remains in $C_{0,n}$. Direct substitution shows that this product law is

$$Q_\delta(z \mid x) = P_i(z)^\delta \frac{h(y)}{h(x)}. \quad (91)$$

Thus along every surviving prefix w in this component,

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h(x_T)}{h(x_0)}. \quad (92)$$

The endpoint term is bounded. The mass-distribution argument proves attainment and the Hausdorff lower bound. For the upper and pointwise bounds, note that along any infinite run the pair (a, b) changes at most $2n$ times and is eventually fixed. Applying the same componentwise coding argument on the final $C_{a,b}$, whose pressure root is at most δ , proves the effective and Hausdorff upper bounds for all nonterminating tapes.

It remains to prove the state bound. For each $S \subseteq \{1, \dots, n\}$, let $x_0^S = 0$ and $x_i^S = 1$ exactly for $i \in S$. Under the least-index rule, repeated outcome 00 deletes the leftmost one; this distinguishes sets of different sizes. If $S = \{s_1 < \dots < s_k\}$ and $T = \{t_1 < \dots < t_k\}$ are distinct, choose $r < k$ with unequal suffix sums. After r outcomes 00, repeated outcome 10 decreases by one the inversion count

$$I(x) = \#\{(i, j) : i < j, x_i = 0, x_j = 1\}.$$

The two counts are

$$\sum_{j=r+1}^k s_j - \binom{k-r}{2} \quad \text{and} \quad \sum_{j=r+1}^k t_j - \binom{k-r}{2},$$

so one run halts first under a common continuation. The 2^n assignments x^S are therefore pairwise continuation-distinct. $\square$

Proof of Corollary 5.8. The characteristic polynomials of the leading principal submatrices satisfy the three-term recurrence

$$D_{j+1}(\lambda) = \lambda D_j(\lambda) - (K_\theta)_{j-1,j}^2 D_{j-1}(\lambda).$$

It costs $O(n)$ operations to evaluate the Sturm sequence at a trial λ , and $O(\log(1/\varepsilon))$ trials isolate the largest root in a fixed bounded interval. The remaining claims follow from Theorem 5.7 and the strict monotonicity of $\rho(K_\theta)$. $\square$

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